

Lecture(18)

Chapter(5)

Counting

- Permutations.
- Combinations.

Permutations

- A **Permutation** is an arrangement of objects (n) in a particular order.

Notice, **ORDER MATTERS!**

- In other words, a permutation is an arrangement of objects, **without repetition, and order being important.**
- The number of permutations of n items taken r at a time is denoted by $P(n, r)$.

Example 1: List all permutations of the letters ABCD

ABCD	BACD	CABD	DABC
ABDC	BADC	CADB	DACB
ACBD	BCAD	CBAD	DBAC
ACDB	BCDA	CBDA	DBCA
ADBC	BDAC	CDAB	DCAB
ADCB	BDCA	CDBA	DCBA

Permutations

Example 2: List all three letter permutations of the letters in the word HAND

HAN	AHN	NHD	DHA
HNA	ANH	NDH	DAH
HAD	AHD	NAH	DAN
HDA	ADH	NHA	DNA
HND	AND	NAD	DHN
HDN	ADN	NDA	DNH

Now, if you didn't actually need a listing of all the permutations, you could use the formula for the number of permutations in the next slide.

Permutations

- To find the number of Permutations of n items chosen r at a time, you can use the formula

$$P(n, r) = \frac{n(n-1)(n-2)\dots(n-r+1)(n-r)!}{(n-r)!} = \frac{n!}{(n-r)!} \quad \text{where } 0 \leq r \leq n .$$

- The n value is the total number of objects to chose from. The r is the number of objects your actually using.
- In the special case in which $r = n$, we have $P(n, n) = n!$

Corollary: There are $n!$ permutations of n objects (taken all at time).

For example, there are $3! = 6$ permutations of the three letters a , b , and c . That is

Permutations

The number of ways to arrange the letters *ABC*:

	_____	_____	_____
Number of choices for first blank?	3	_____	_____
Number of choices for second blank?	3	2	_____
Number of choices for third blank?	3	2	1

$$3*2*1 = 6$$

$$3! = 3*2*1 = 6$$

ABC ACB BAC BCA CAB CBA

Permutations

Practice :

A combination lock will open when the right choice of three numbers (from 1 to 30, inclusive) is selected. How many different lock combinations are possible assuming no number is repeated?

Answer Now

Permutations

Practice:

A combination lock will open when the right choice of three numbers (from 1 to 30, inclusive) is selected. How many different lock combinations are possible assuming no number is repeated?

$${}_{30}P_3 = \frac{30!}{(30-3)!} = \frac{30!}{27!} = 30 * 29 * 28 = 24360$$

Permutations

Practice:

From a club of 24 members, a President, Vice President, Secretary, Treasurer and Historian are to be elected. In how many ways can the offices be filled?

Answer Now

Permutations

Practice:

From a club of 24 members, a President, Vice President, Secretary, Treasurer and Historian are to be elected. In how many ways can the offices be filled?

$${}_{24}P_5 = \frac{24!}{(24-5)!} = \frac{24!}{19!} =$$

$$24 * 23 * 22 * 21 * 20 = 5,100,480$$

More Examples

- (1) Find the number of ways that a party of seven persons can arrange themselves in a row of seven chairs.

Solution: The seven persons can arrange themselves in a row in $7.6.5.4.3.2.1=7!$ ways.

- (2) How many different ways are there to select 4 different players from 10 players on a team to play four tennis matches. Where the matches are ordered?

Solution: $P(10, 4) = 10.9.8.7 = 5040$.

- (2) Find n if, $P(n,2)=72$.

Solution: $P(n,2) = n(n-1)$

Hence $n(n-1) = 72$ or

$$(n-9)(n+8)=0$$

Since n must be positive, the only answer is $n=9$.

- (3) Find n if $3P(n, 2) + 27 = P(3n, 2)$.

- (4) How many permutations of $\{a, b, c, d, e, f, g\}$ end with a .

Solution: 720

Note that the set has 7 elements

The last character must be a

The rest can be in any order

Thus, we want a 6-permutation on the set $\{b, c, d, e, f, g\}$

Distinguishable Permutations

- Consider all the permutations of the letters in the word **BOB**. Since there are three letters, there should be $3! = 6$ different permutations.
- Those permutations are **BOB**, **BBO**, **OBB**, **OBB**, **BBO**, and **BOB**. Now, while there are six permutations, some of them are **indistinguishable** from each other.
- If you look at the permutations that are **distinguishable**, you only have three **BOB**, **OBB**, and **BBO**.
- To find the number of distinguishable permutations, take the total number of letters factorial divide by the frequency of each letter factorial.

$$\frac{N!}{(n_1!)(n_2!)\dots(n_k!)} \quad \text{Where } n_1 + n_2 + \dots + n_k = N$$

Distinguishable Permutations

Examples of distinguishable permutations

(1) Find the number of distinct permutation that can be formed from all letters of the word "BENZENE".

Solution:

$$P(7;1,3,2) = \frac{7!}{1!3!2!} = 420$$

(2) Find the number of distinguishable permutations of the letters in the word MISSISSIPPI

$$P(11;1,4,4,2) = \frac{11!}{1!4!4!2!} = 34650$$

(3) MATHMATICS

(4) ALABAMA

Combinations

- A **Combination** is an arrangement of items in which order does not matter.

ORDER DOES NOT MATTER!

- In other words, a combination is an arrangement of objects, without repetition, and order not being important.

Combinations

Example 1: List all permutations of the letters ABCD in group of 3.

There are only four combinations (**ABC**, **ABD**, **ACD**, and **BCD**). Listed below each of those combinations are the six permutations that are equivalent as combinations.

ABC	ABD	ACD	BCD
ABC	ABD	ACD	BCD
ACB	ADB	ADC	BDC
BAC	BAD	CAD	CBD
BCA	BDA	CDA	CDB
CAB	DAB	DAC	DBC
CBA	DBA	DCA	DCB

- Since the order does not matter in combinations, there are fewer combinations than permutations. The combinations are a "subset" of the permutations.

Combinations

- The number of combinations of n objects taken r at a time is denoted by $C(n,r)$ or $\binom{n}{r}$
- To find the number of Combinations of n items chosen r at a time, you can use the formula
- The n and r in the formula stand for the total number of objects to choose from and the number of objects in the arrangement, respectively.

$${}_n C_r = \frac{n!}{r!(n-r)!} \quad \text{where } 0 \leq r \leq n.$$

Combinations

Practice:

To play a particular card game, each player is dealt five cards from a standard deck of 52 cards. How many different hands are possible?

Answer Now

Combinations

Practice:

To play a particular card game, each player is dealt five cards from a standard deck of 52 cards. How many different hands are possible?

$${}_{52}C_5 = \frac{52!}{5!(52-5)!} = \frac{52!}{5!47!} =$$
$$\frac{52 * 51 * 50 * 49 * 48}{5 * 4 * 3 * 2 * 1} = 2,598,960$$

Combinations

Practice:

A student must answer 3 out of 5 essay questions on a test. In how many different ways can the student select the questions?

Answer Now

Combinations

Practice:

A student must answer 3 out of 5 essay questions on a test. In how many different ways can the student select the questions?

$${}_5C_3 = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5*4}{2*1} = 10$$

Combinations

Practice:

A basketball team consists of two centers, five forwards, and four guards. In how many ways can the coach select a starting line up of one center, two forwards, and two guards?

Answer Now

Combinations

Practice:

A basketball team consists of two centers, five forwards, and four guards. In how many ways can the coach select a starting line up of one center, two forwards, and two guards?

Center:

$${}_2C_1 = \frac{2!}{1!1!} = 2$$

Forwards:

$${}_5C_2 = \frac{5!}{2!3!} = \frac{5*4}{2*1} = 10$$

Guards:

$${}_4C_2 = \frac{4!}{2!2!} = \frac{4*3}{2*1} = 6$$

$${}_2C_1 * {}_5C_2 * {}_4C_2$$

Thus, the number of ways to select the starting line up is $2*10*6 = 120$.

More Examples

(1) How many committees of three can be formed from eight people?

Solution: Number of committees that can be formed is $C(8,3) = \binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$

(2) A farmer buys 3 cows, 2 pigs and 4 hens from a man who has 6 cows, 5 pigs and 8 hens. How many choices does the farmer have?

Solution: The farmer can choose the cows in $\binom{6}{3}$ ways, the pigs in $\binom{5}{2}$ ways, and the hens in $\binom{8}{4}$ ways.

Hence altogether he can choose the animals in $\binom{6}{3} \binom{5}{2} \binom{8}{4} = 20 \cdot 10 \cdot 70 = 14000$ ways.

(3) How many committees of five with a given chairperson can be selected from 12 persons?

Solution: The chairperson can be chosen in 12 ways and, following this, the other four on the committee can be chosen from the eleven remaining in $\binom{11}{4}$

ways. Thus there are $12 \binom{11}{4} = 12 \cdot 330 = 3960$ such committees.

(4) How many ways are there to select 5 players from a 10-member tennis team to make a trip to a match at another school?

Solution: $C(10,5) = 10!/(5!5!) = 252$.

(5) How many ways are there to select a committee to develop a discrete mathematics course at KFU if the committee is to consist of 3 faculty members from the mathematics department and 4 from the computer science department, if there are 9 faculty members of the mathematics department and 11 of the computer science department

Solution: The number of ways to select the committee is:

$$C(9,3) \cdot C(11,4) = \frac{9!}{3!6!} \cdot \frac{11!}{4!7!} = 84 \cdot 330 = 27720.$$

Lecture(19)

Chapter(5)

Counting

- The Pigeonhole Principle.
- The Inclusion-Exclusion Principle.
- Ordered and Unordered Partitions.